* 1 point

Question 1

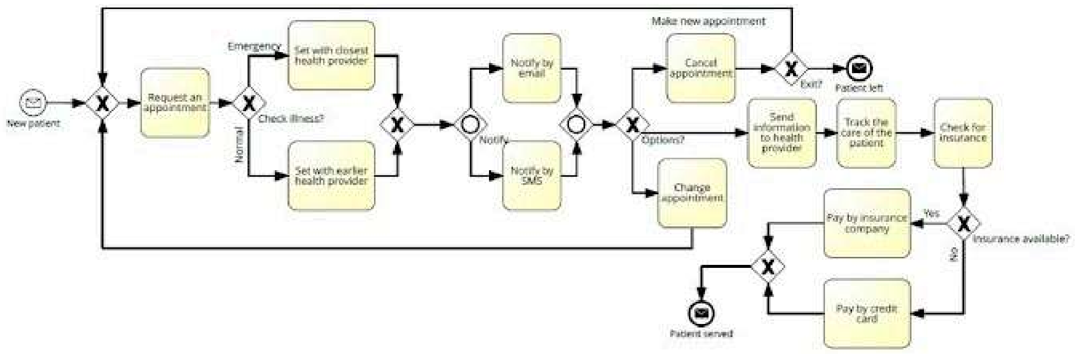
Consider the following BPM.

Given the following BPMN model

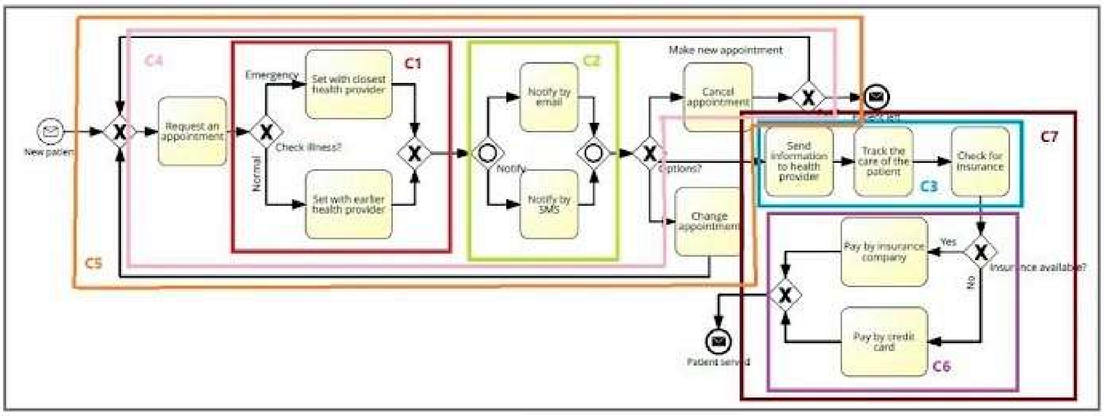
Answer all the following Questions (from question 1 to 6)

```
<Switch>
      <Case condition="Yes">
     <invoke name ="set with earlier Health provider"/>
       </case>
      <case condition="No">
      <invoke name ="set with closest Health provider "/>
     </case>
  </Switch>
```

This BPEL represents which component?



BPMN after folding it into components



- C3
- C2
- C4
- C1

Question 2

* 1 point

The following is the mapping of component (C3), but it has an error. Which part contains the error?

```
< flow >
  <invoke name =" send        the information to the health provider" />
   <invoke name =" track the care of the patient"/>
   <invoke name =" check the insurance"/>
</ flow >
```

- (selected) The <flow> tag should be <sequence>
- <Invoke> should be <receive>
- The <flow> tag should be <switch>
- None of the above is true

Question 3 * 1 point

The OR gateway in component (C2) should be represented in BPEL as:

```
<flow>
<Case condition="email">
<invoke name ="notify by email" />
</case>
<Case condition= "sms">
<invoke name ="notify by sms"/>
</case>
<Case condition="email" and "sms">
<invoke name ="notify by email"/>
<invoke name ="notify by sms"/>
</case>
</flow>
```

○ Option 1

```
<switch>
<if condition="email">
<invoke name ="notify by email" />
</if>
<if condition= "sms">
<invoke name ="notify by sms"/>
</if>
<if condition="email" and "sms">
<invoke name ="notify by email"/>
<invoke name ="notify by sms"/>
</if>
</switch>
```

○ Option 2

```
<flow>
< if condition="email">
<invoke name ="notify by email" />
</ if >
< if condition= "sms">
<invoke name ="notify by sms"/>
</ if >
< if condition="email" and "sms">
<invoke name ="notify by email"/>
<invoke name ="notify by sms"/>
</ if >
</flow>
```

○ Option 3

```
< sequence >
< case condition="email">
<invoke name ="notify by email" />
</ case >
< case condition= "sms">
<invoke name ="notify by sms"/>
</ case >
< case condition="email" and "sms">
<invoke name ="notify by email"/>
<invoke name ="notify by sms"/>
</ case >
</ sequence >
```

○ Option 4

◉ None of the above

Question 4 * 1 point

```
< sequence >

<invoke name ="request an appointment"/>

Mapping C1

Mapping C2

<invoke name ="cancel appoint "/>

< while condition = "make a new appointment">

<sequence>

<invoke name ="request an appointment"/>

Mapping C1

Mapping C2

<invoke name ="cancel appointment "/>

    </sequence>

</while>

< /sequence >
```

This BPEL code is the mapping of C5?

○ True

◉ False

Question 5 * 1 point

```
< sequence >

Mapping  C1

Mapping  C2

 <invoke name ="change appointment" />

[Missing part 1]

         <sequence>

           Mapping  C1

           Mapping  C2

          <invoke name ="change appointment" />

      </sequence>



</sequence>
```

This BPEL represents the mapping of component (C5), but there is a missing part 1

Choose the correct one for this missing part:

- (selected) <while condition =" change the appointment > and </while>
- <forEach condition =" change the appointment "> and </forEach>
- <if condition =" change the appointment "> and </if>
- <repeatUntil condition =" change the appointment "> and <repeatUntil>

Question 6 Bonus (3 marks) 3 points

Write the full BPEL code for the above BPMN

Please attach file as pdf or word only

Upload 1 supported file: PDF or document. Max 100 MB.

Add file

The second part Petri-net

Question 7

* 1 point

To translate a BPMN business process model to a Petri net, a transformation is provided where elements of the BPMN model are mapped to Petri net fragments(subnets).

Which is the correct Petri net transformation for an **intermediate event**?

- ○ a place with several incoming transitions.
- ○ a place with several outgoing transitions.
- ○ a transition with one output place that has no outgoing arcs.
- ○ a transition with one input place that carries a token but has no incoming arc.
- ○ a transition with several output places.
- ○ a transition with several input places.
- ◉ a transition with one input place and one output place.

Question 8

* 1 point

To translate a BPMN business process model to a Petri net, a transformation is provided where elements of the BPMN model are mapped to Petri net fragments(subnets).

Which is the correct Petri net transformation for a parallel split gateway?

- ○ a place with several incoming transitions.
- ○ a transition with one output place that has no outgoing arcs.
- ○ a place with several outgoing transitions. 3
- ○ a transition with one input place that carries a token but has no incoming arc.
- ◉ a transition with several output places.
- ○ a transition with several input places
- ○ a transition with one input place and one output place

Question 9

* 1 point

To translate a BPMN business process model into a Petri net, a transformation is provided that maps elements of the BPMN model to Petri net fragments (subnets).

Which is the correct Petri net transformation for a parallel join gateway?

- ○ a place with several incoming transitions
- ○ a place with several outgoing transitions.
- ○ a transition with one output place that has no outgoing arcs.
- ○ a transition with one input place that carries a token but has a no incoming arc.
- ○ a transition with several output places.
- ◉ a transition with several input places
- ○ a transition with one input place and one output place.

Question 10

* 1 point

To translate a BPMN business process model to a Petri net, a transformation is provided where elements of the BPMN model are mapped to Petri net fragments(subnets).

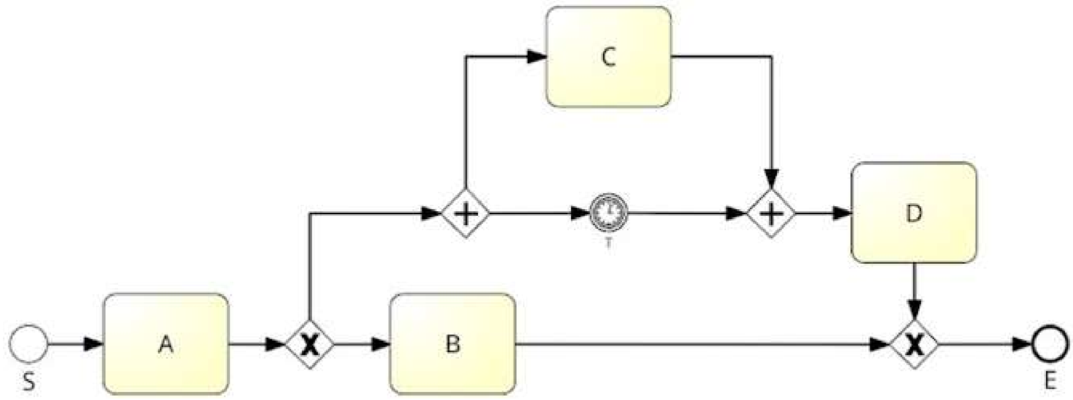
Which is the correct Petri net transformation for an exclusive join gateway?

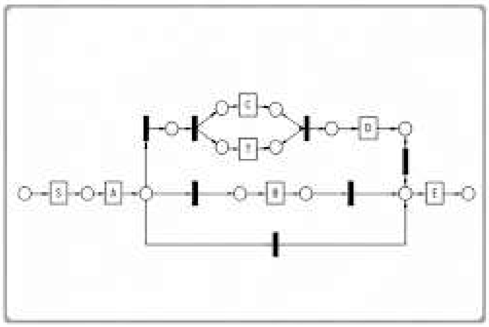
- ◉ a place with several incoming transitions.
- ○ a place with several outgoing transitions.
- ○ a transition with one output place that has no outgoing arcs.
- ○ a transition with one input place that carries a token but has no incoming arc.
- ○ a transition with several output places.
- ○ a transition with several input places.
- ○ a transition with one input place and one output place.

Question 11 * 1 point

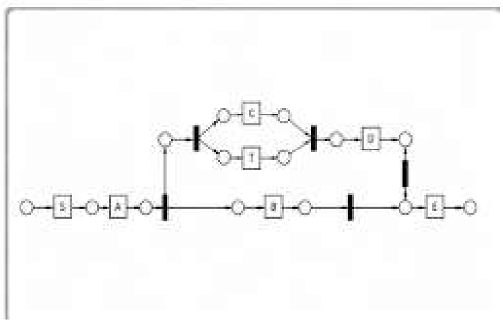
Given the following process model:

Which of the following Petri nets is the correct transformation of the process model into a Petri net?

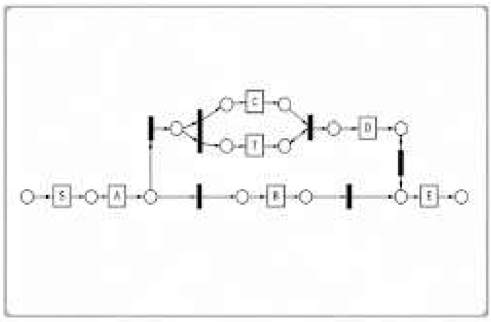




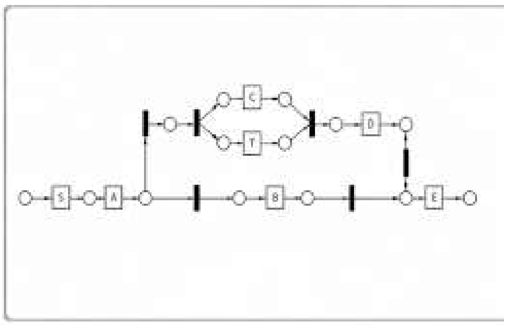
Option 1



Option 2



Option 3



Option 4

Send me a copy of my responses.